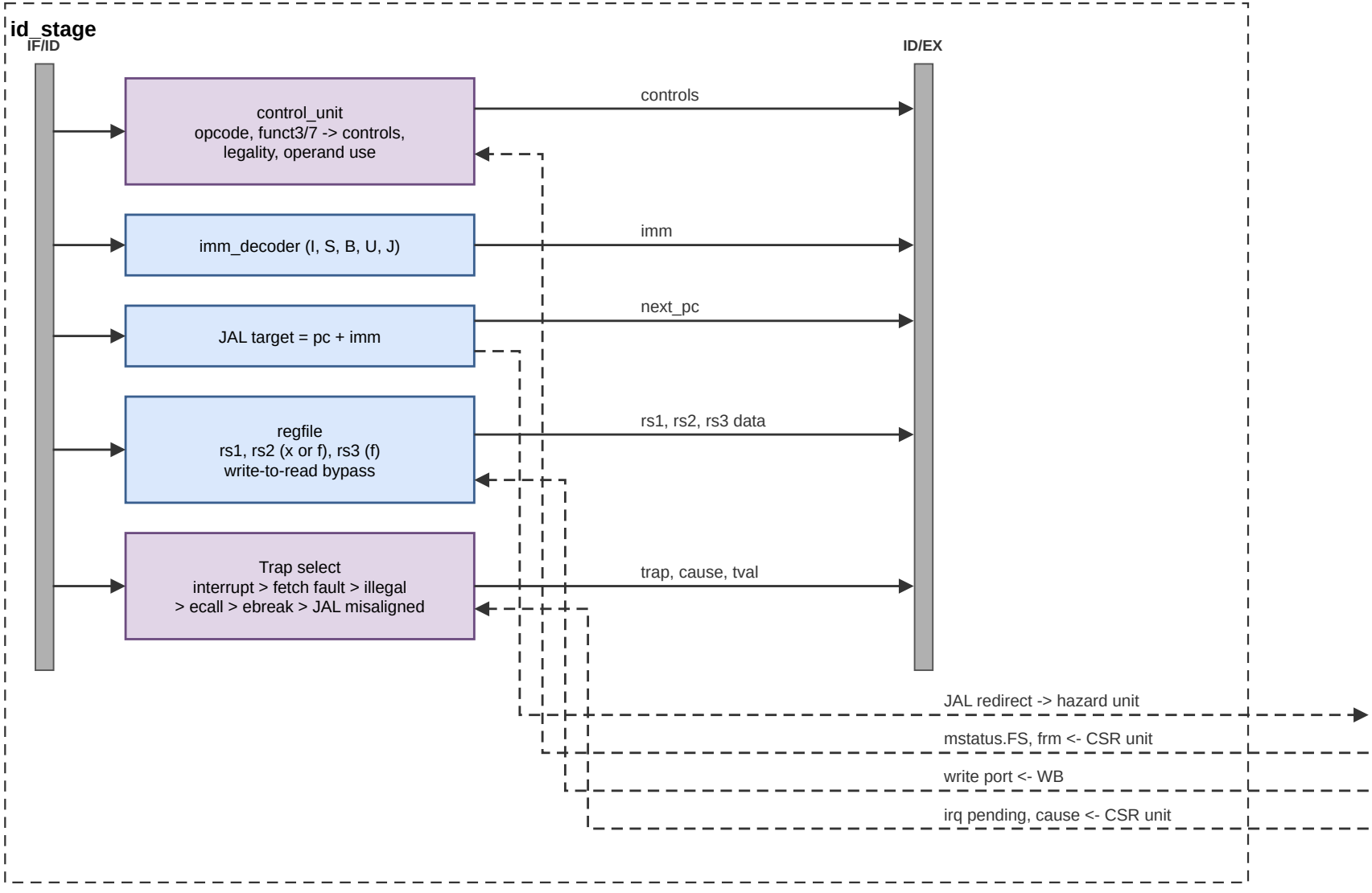
id_stage
IF/ID
ID/EX
control_unit
opcode, funct3/7 -> controls,
legality, operand use
controls
imm_decoder (I, S, B, U, J)
imm
JAL target = pc + imm
next_pc
regfile
rs1, rs2 (x or f), rs3 (f)
write-to-read bypass
rs1, rs2, rs3 data
Trap select
interrupt > fetch fault > illegal
> ecall > ebreak > JAL misaligned
trap, cause, tval
JAL redirect -> hazard unit
mstatus.FS, frm <- CSR unit
write port <- WB
irq pending, cause <- CSR unit